Verification Lab 5 - Formal

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1 Introduction

Now that you have attained godlike status with SystemVerilog and UVM, let's switch gears to formal verification. Here, we're going to get our hands dirty with the VC Formal solution from Synopsys, which includes a comprehensive set of formal applications, as follows:

- Property Verification (FPV).
- Automatic Extracted Properties (AEP).
- Coverage Analyzer (FCA).
- Connectivity Checking (CC).
- Sequential Equivalence Checking (SEQ).
- Register Verification (FRV).
- X-Propagation Verification (FXP).
- Testbench Analyzer (FTA).
- Regression Mode Accelerator (RMA).
- Datapath Validation (DPV).
- Functional Safety (FuSa).
- Assertion IPs (AIP).

If you are curious about the particulars, you can scratch that itch by checking out the VC Formal datasheet here: VC Formal Datasheet

Alright, let's get to it then. For this lab, you're going to use the lab material that comes shipped with VC Formal itself. To access these, you need to make your way to this directory:

```
/usr/local-eit/cad2/synopsys/vcf24/vc\_static/V-2023.12/doc/vcst/examples
```

There, you can find all sorts of labs for the different formal applications supported by VC Formal. But, for now, we're only going to do the one contained in the `FPV` directory; however, you're more than welcome to check out the rest, if it takes your fancy.

Copy the contents of the `FPV/FPV` directory to your student drive, or wherever you can modify the contents. Then navigate to it in the terminal and run:

```
csh
```

This command makes a switch form *bash* to *C shell*. Afterwards, you can source the setup file:

```
source /usr/local-eit/cad2/synopsys/vcf24/setup
```

Now with that out of the way, open the `README\_VCFormal\_FPV.pdf` document contained in the directory and follow its instructions. Skip the environment setup in the `Prepare your Environment` section, as you've already made the tool available by sourcing the setup file. Form here on out just follow the PDF. As I said before, there are also other labs that you can try out on your own. Now, go get 'em tiger(s).